

# STRING & RACKET FINDER

## The Engine Explained

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How the scoring works, what makes it different, and why results sometimes surprise you.

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### 01 — ORIGIN

## Built from scratch. No third-party data.

The String & Racket Finder's scoring engine was developed entirely in-house by A<sup>3</sup> Agentur. It uses no external rating databases, no crowd-sourced review averages, no manufacturer-supplied performance claims, and no affiliate ranking logic. Every score is computed from the physical properties of the equipment matched against the physical demands of the player's game.

There is no comparable free tool with this methodology. The closest analogues — Tennis Warehouse's racket filter, Tennis-Point's string browser, and racketfinder.com — operate on a fundamentally different principle: they let players browse and filter by specification. The Finder does the opposite. It asks what kind of player you are and computes what the equipment will do for you specifically.

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### 02 — TWO MODES

## Scored Mode vs. Pure Physics Mode

### Scored Mode — Default

Each racket frame is evaluated across 11 performance axes; each string bed across 8 axes. Scores are derived from a combination of physical specifications — stiffness, swingweight, beam width, head size, string pattern, balance, gauge, cross-section profile, material class — and, where available, cross-referenced laboratory figures published for the string itself: dynamic stiffness, tension loss curves, and energy return.

The engine applies a hierarchy of inputs. For a string bed it anchors the range on a reference stiffness and positions every other string relative to that anchor: stiffer strings gain in control and precision and lose in arm friendliness and feel; softer strings gain in power and comfort and lose in control. Cross-section profile modulates spin access — the more edges a string presents to the ball, the more bite it is credited with. Tension loss adjusts longevity; energy return adjusts power.

### Pure Physics Mode

A stricter, more transparent mode. Only directly measurable physical quantities are used — no estimated nudges, no inferred axes. Axes for which no physical measurement exists are left empty rather than estimated. For frames: Serve, Smash, and Slice/touch are left empty. For string beds: Precision and Sound are empty. An empty cell is more honest than an invented one.

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### 03 — INTEGRITY

## The Honesty Rules

## "No medals without receipts."

— Estimated values are capped at 9. A score of 10 cannot be assigned by the engine. A 10 must be earned on court — it only appears when a human has played the equipment and confirmed the rating. Every engine-derived value carries a visible dagger symbol (†) marking it as an estimate.

— Identical inputs produce identical outputs — labelled. If two string beds share the same material class, profile, gauge, and lab row, they score identically. Rather than introduce artificial variation, the engine flags both as "twin est." with a chip pointing to the mirror entry. Invented differences would break the engine's core promise.

— No manufacturer influence. No brand pays to appear in results. No string is seeded higher because of a commercial relationship. Every brand scores from the same inputs on the same scale.

### 04 — AXES

## 11 Racket Axes · 8 String Axes

### Racket

|    |                   |                                                 |
|----|-------------------|-------------------------------------------------|
| 1  | Control (flat)    | Performance on driven, flat groundstrokes       |
| 2  | Precision         | Placement accuracy — distinct from Control      |
| 3  | Spin              | Access to topspin and rotation                  |
| 4  | Power             | Frame's own energy contribution                 |
| 5  | Feel / Comfort    | Feedback quality and arm friendliness           |
| 6  | Stability         | Torsional resistance against incoming pace      |
| 7  | Maneuverability   | Swing ease and reaction speed                   |
| 8  | Serve             | Access to pace, kick, and slice on first strike |
| 9  | Smash & Overhead  | Overhead finishing power                        |
| 10 | Slice / Touch     | Drop shots, carving, underspin                  |
| 11 | Return / Blocking | Absorbing incoming pace                         |

Difficulty is displayed separately as an A<sup>3</sup>-logo triangle. It is not part of the ranking. A score of 10 means the frame is the most demanding in the database — not the best.

### String

|   |                  |                                                    |
|---|------------------|----------------------------------------------------|
| 1 | Spin             | Rotation access from profile and snapback geometry |
| 2 | Power            | Energy return from the bed                         |
| 3 | Control (flat)   | Precision on flat, driven shots                    |
| 4 | Precision        | Placement repeatability                            |
| 5 | Feel             | Touch and feedback quality                         |
| 6 | Sound            | Acoustic signature                                 |
| 7 | Longevity        | Tension maintenance over time                      |
| 8 | Arm Friendliness | Shock and vibration load on the arm                |

## The Player Profile

The Finder asks three questions before it touches the equipment database.

### Playing Level

Not a ranking number — a self-assessed skill description tied to the NTRP scale. The Pro Mode includes the Player Bench: a structured self-evaluation across 10 playing axes (Forehand, Backhand, First Serve, Second Serve, Return, Volley & Net, Smash & Overhead, Slice/Touch, Movement, Point Construction Under Pressure). Each axis has pass-mark drills producing an NTRP bracket with UTR range displayed alongside.

Two integrity rules apply: estimated axes are capped at 8 — a 4.5 must be earned on court — and a Pressure Gate stops the final rating running ahead of the Second Serve and Point Construction scores, the two axes that most reliably separate levels.

### Playing Priorities

Eleven weight sliders let players dial the relative importance of each axis. Control (flat) defaults to 10; everything else defaults to 5. Moving a slider changes the ranking. A player who slides Spin to 10 and Control to 3 will see a completely different shortlist than one who does the reverse — on the same set of strings.

### Amplify Your Strengths — or Cover Your Weaknesses

A Compensate ⇔ Amplify slider sets the stance. This is where results become genuinely surprising.

## Amplify vs. Cover — Why Results Surprise You

### The Amplify path

A player who is already a strong spin producer sets Spin to 10. The engine weights spin heavily and returns string beds that maximise what this player already does well. Results are predictable: shaped polys, sharp profiles, high snapback. The shortlist feels familiar.

### The Cover path

A player with a slow swing and arm sensitivity sets Arm Friendliness to 10, Spin to 6. The engine is now searching for a string that can generate relative spin improvement without arm load — a narrow intersection. Results may include strings this player would never have searched for, because their name or category does not signal "spin string." A softer multifilament with a shaped profile. A natural gut hybrid. A co-poly at the low end of the stiffness band with a shaped cross-section.

The string may appear in the shortlist because it is the softest string in the database that still has a shaped profile — a property combination that no keyword search can surface, because no retailer tags strings that way.

### Why the same string appears from many different searches

A string bed contributes to 8 axes simultaneously. A string with high tension maintenance, soft feel, and a shaped profile will surface in searches prioritising arm friendliness, in searches prioritising longevity, in a search for soft strings with some spin, in a cover-weakness search by a heavy hitter with arm problems, and in an amplify search by a player who already relies on feel.

The same string appears from five different player profiles, for five different reasons. A player browsing by category would find it under one heading, if they knew to look. The engine finds it under all five.

**The shortlist is not "the best strings for spin" or "the best strings for arm health." It is "the best string for this specific combination of priorities at this specific level" — a combination that often has no named category.**

## 07 — COMPARISON

### How This Compares to Other Tools

|                         | String & Racket Finder                  | Tennis Warehouse             | Tennis-Point         | Racketfinder.com   |
|-------------------------|-----------------------------------------|------------------------------|----------------------|--------------------|
| Search method           | Physics engine + player profile         | Filter by spec               | Filter by spec/brand | Filter by category |
| Player input            | Level + priorities + stance             | Level (broad)                | None                 | None               |
| Scoring axes            | 11 racket / 8 string                    | Star rating (user-generated) | None                 | None               |
| Frame×string calibrator | Yes                                     | No                           | No                   | No                 |
| 3rd-party data          | Published lab figures, cross-referenced | Review aggregation           | Mfr specs            | Mfr specs          |
| Paid placement          | None                                    | Advertising present          | Advertising present  | Not disclosed      |
| Mode switching          | Scored ⇄ Pure Physics                   | None                         | None                 | None               |
| Weakness coverage       | Yes — Compensate ⇄ Amplify              | No                           | No                   | No                 |

Tennis Warehouse's search is the most sophisticated of the three comparators — structured filters, expert reviews, level-based recommendations. But it returns lists to browse, not a match. A player who does not already know what they are looking for has no way to navigate to the right string from a list of 396 beds.

The Finder eliminates the list entirely. It returns four setups: Gold, Silver, Bronze, Budget. Each carries a why-line explaining which axes drove the recommendation and what the frame-string pairing contributes. If the top recommendation was gated — the highest raw-math score excluded because it conflicted with another priority — the engine names the gated string and explains why.

**Transparency is not a feature. It is the architecture.**

String & Racket Finder is free, independently operated, and has no commercial relationships that affect scoring. · [stringandracketfinder.com](https://stringandracketfinder.com) · A<sup>3</sup> Agentur Munich · [www.a3agentur.com](https://www.a3agentur.com)